

AI-Driven Peptide Binder Design Against the Nipah Virus Receptor-Binding Protein

Protein Design Challenge: Foundations of Protein Language & AI



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1. INTRODUCTION

Nipah virus (NiV) is a highly lethal zoonotic paramyxovirus first identified in Malaysia in 1999. With case fatality rates ranging from 40% to 75%, it is classified as a priority pathogen by the World Health Organization due to its epidemic potential and the current absence of approved therapeutics or vaccines for human use.

The Nipah virus attaches to host cells primarily through its receptor-binding protein (RBP), also referred to as the attachment glycoprotein (G protein). This protein mediates viral entry by recognising and binding to host cell surface receptors, namely Ephrin-B2 and Ephrin-B3, which are expressed in neurons and vascular endothelial cells. Blocking this interaction is considered one of the most promising therapeutic strategies against NiV infection.

The structural reference used in this study is PDB entry 2VSM, which is the experimentally resolved crystal structure of the Nipah virus G protein. This structure, determined by X-ray crystallography, provides an accurate atomic-resolution map of the viral attachment surface. The target sequence used for structure prediction and hotspot analysis in this project is a 602-residue protein beginning with MPAENKKVRFENTT and ending with KLFVAVKIPEQCT, spanning the full-length receptor-binding domain of the NiV G protein.

This project applies a state-of-the-art AI-driven protein design pipeline to generate novel peptide binders targeting the Nipah virus RBP. The pipeline integrates deep learning models for structure prediction, diffusion-based backbone generation, inverse folding for sequence design, and structure-based validation — collectively representing the current frontier of computational drug design.

2. OBJECTIVES

- To predict the 3D structure of the Nipah virus receptor-binding protein from its primary sequence using AI-based structure prediction and identify druggable binding pockets on its surface.
- To design novel peptide binder backbones and sequences targeting the identified hotspot region of the Nipah virus G protein using generative AI models (RFDiffusion and ProteinMPNN).
- To validate and rank the designed peptide sequences based on structural confidence metrics, and select the top 10 candidates with the highest predicted binding and folding quality scores.

3. METHODOLOGY

The entire pipeline was executed using a combination of online bioinformatics platforms and Google Colab notebooks provided as part of the Omics Summer School 2026 hands-on curriculum. The workflow consisted of six sequential steps, summarised in the pipeline table below.

Step	Tool	Input	Output
1	ESMFold2 (Biohub)	Target sequence	3D structure PDB file
2	fpocket (RPBS Server)	PDB file	Binding pockets + hotspot residues
3	RFdiffusion (Colab)	PDB + hotspot residues	Peptide backbone PDB files
4	ProteinMPNN (Colab)	Backbone PDB files	32 candidate sequences (FASTA)
5	AlphaFold (Colab)	Candidate sequences	pLDDT, pTM, PAE, RMSD scores
6	Manual ranking	Score table	Top 10 peptide binder sequences

3.1 Structure Prediction

The target sequence (602 residues) was submitted to the ESMFold2 model via the Biohub platform (biohub.ai/tools/fold). The model returned a predicted structure with a Mean pLDDT of 0.67 and pTM of 0.65. Given the inherent flexibility of viral glycoproteins, these confidence values were deemed sufficient for downstream analysis. The output was downloaded in mmCIF format and converted to PDB format for compatibility with subsequent tools.

3.2 Binding Pocket Detection

The predicted PDB file was submitted to the fpocket server hosted at the RPBS web portal (mobyale2.rpbs.univ-paris-diderot.fr). A total of 22 binding pockets were detected. Pocket 0 was selected as the primary hotspot target based on its superior druggability metrics: volume of 996 Å³, depth of 19.54 Å, and a drug score of 0.79. Hotspot residues identified from Pocket 0 were A176, A191, A197, and A199.

3.3 Backbone Generation via RFdiffusion

The RFdiffusion Colab notebook was used to generate novel peptide backbones. The real crystal structure 2VSM was used as the input to ensure atomic-level accuracy. Key parameters included: contigs length of 100, 50 diffusion iterations, 8 designs, and hotspot residues A176, A191, A197, A199. RFdiffusion produced 8 de novo peptide backbone PDB files targeting the identified binding pocket.

3.4 Sequence Design via ProteinMPNN

The generated backbone PDB files were passed to ProteinMPNN (integrated within the Colab notebook). Four sequences were generated per backbone design, resulting in a total of 32 candidate peptide sequences. Parameters used: initial_guess = True, num_recycles = 3, mpnn_sampling_temp = 0.25, rm_aa = C.

3.5 Validation and Scoring

All 32 sequences were validated using AlphaFold2 (integrated within the pipeline notebook), which computed four structural quality metrics for each candidate: pLDDT (predicted local distance difference test), pTM (predicted template modelling score), PAE (predicted aligned error), and RMSD (root mean square deviation). Results were exported as a CSV file and ranked in descending order of pLDDT to identify the top 10 binder candidates.

4. RESULTS

4.1 Structure Prediction Output

ESMFold2 successfully predicted a folded 3D structure for the Nipah virus RBP from the raw sequence. The predicted structure showed a globular architecture with clearly defined helical and beta-sheet regions consistent with the known paramyxovirus attachment glycoprotein fold. The pLDDT score of 0.67 indicated moderate-to-good confidence, with higher confidence concentrated in the structural core. The pTM score of 0.65 confirmed a plausible overall fold topology.

4.2 Hotspot Identification

fpocket detected 22 pockets on the protein surface. Pocket 0 emerged as the most promising binding site, exhibiting the highest volume (996 Å³), deepest cavity (19.54 Å), and the highest drug score (0.79) among all detected pockets. The selected hotspot residues — Gln176, Glu191, Arg197, and Ser199 — are located within a central depression on the protein surface, consistent with known receptor interaction sites in paramyxovirus G proteins.

4.3 Top 10 Peptide Binder Candidates

Following ProteinMPNN sequence generation and AlphaFold2 validation, the 32 candidate sequences were ranked by pLDDT. The top 10 sequences are presented in Table 2 below. All top candidates achieved pLDDT scores above 0.91, indicating high structural confidence. Design 26 ranked first with the highest pLDDT of 0.962 and a pTM of 0.826, suggesting both strong local folding accuracy and reliable global topology prediction.

Table 2: Top 10 Peptide Binder Candidates Ranked by pLDDT Score

Rank	Design	pLDDT	pTM	PAE	RMSD
1	26	0.962	0.826	2.582	0.557
2	29	0.940	0.737	3.091	0.598
3	11	0.940	0.808	2.329	0.833
4	23	0.937	0.756	3.088	1.040
5	24	0.937	0.764	3.098	0.571

Rank	Design	pLDDT	pTM	PAE	RMSD
6	31	0.936	0.750	2.690	0.649
7	30	0.933	0.694	6.326	2.097
8	10	0.926	0.735	4.500	1.113
9	1	0.930	0.733	3.553	1.228
10	27	0.915	0.729	3.619	0.771

The low PAE values observed in the top-ranking designs (particularly Design 26, PAE = 2.58) indicate minimal positional uncertainty across residues, further supporting the structural reliability of these candidates. The RMSD values, where low, reflect close agreement with the target backbone geometry generated by RFdiffusion.

5. CONCLUSION

This project successfully demonstrated the application of an end-to-end AI-driven protein design pipeline for the de novo design of peptide binders targeting the Nipah virus receptor-binding protein. Starting from a raw amino acid sequence, the pipeline traversed structure prediction, pocket detection, backbone diffusion, inverse folding, and structure-based validation without requiring any wet-lab experimentation.

The top 10 designed peptides, all achieving pLDDT scores above 0.91, represent structurally confident candidates for further investigation. Design 26 emerged as the lead candidate with a pLDDT of 0.962 and pTM of 0.826, indicating strong predicted folding quality and correct topology. These sequences are predicted to target the primary druggable pocket of the Nipah G protein at hotspot residues 176, 191, 197, and 199.

While computational design provides rapid candidate generation, future work should include molecular docking simulations to assess binding affinity, molecular dynamics simulations to evaluate stability, and ultimately experimental validation through cell-based assays. Nevertheless, the results of this project demonstrate the remarkable potential of generative AI models in accelerating early-stage antiviral drug discovery.

6. TOOLS AND RESOURCES USED

- ESMFold2 — Biohub Platform (biohub.ai/tools/fold): AI-based protein structure prediction from amino acid sequence.
- fpocket — RPBS Web Server (mobyle2.rpbs.univ-paris-diderot.fr): Binding pocket detection and druggability analysis.
- RFdiffusion — Google Colab Notebook: Diffusion-based de novo peptide backbone generation targeting defined hotspot residues.
- ProteinMPNN — Google Colab Notebook (integrated with RFdiffusion): Inverse folding model for sequence design on fixed backbone coordinates.

- AlphaFold2 — Google Colab Notebook (integrated): Structure prediction and scoring of designed sequences (pLDDT, pTM, PAE, RMSD).
- RCSB PDB (rcsb.org): Source of the 2VSM crystal structure of the Nipah virus G protein.
- WPS Office / Microsoft Excel: Tabulation and ranking of candidate sequences by pLDDT score.
- Proteins.Plus / DoGSiteScorer (proteins.plus): Complementary binding site analysis.